A.I Fundamentals

Module 2 Study Guide

Machine Learning Basics

This guide covers every lesson of the module, using only words and figures from the lessons themselves. Work through it after watching the lessons, then use the checklist at the end to test yourself.

Lesson 1: How Machines Learn From Data

In this lesson: Understand training data; Identify features; Build a simple model; Preview next

Key points

Examples from the lesson

- This is the course “A.I Fundamentals”, Module 2 “Machine Learning Basics”, Lesson 1: “How Machines Learn From Data”.
- Imagine we have ten loan applications from members of a St.
- The outcomes we already know are whether the loan was approved (1) or rejected (0).
- Here are the numbers: Applicant 1 earns TT$4,500, owes TT$1,200, has 24 months on the job, has a guarantor, and was approved.
- Applicant 2 earns TT$3,000, owes TT$1,800, has 10 months on the job, no guarantor, and was rejected.
- We continue this way for all ten applicants, building a small table that the machine will study.
- One common rule is the debt-to-income ratio: divide existing debt by monthly income.
- Calculating the ratio for Applicant 1 gives 1,200 ÷ 4,500 ≈ 0.27.
- For Applicant 2 it is 1,800 ÷ 3,000 = 0.60.
- By testing different thresholds on our ten examples, we find that a threshold of 0.35 correctly classifies eight out of ten cases—two mistakes remain.
- We try 0.30, 0.35, 0.40, and see which yields the fewest wrong predictions.
- In this tiny data set, 0.35 gives the best balance, approving six applicants and rejecting four, with only two errors.
- The machine has now learned: if debt-to-income ≤ 0.35, predict loan approval; else predict rejection.
- Next, we will explore the three main families of machine learning—supervised, unsupervised, and reinforcement learning—and see how each uses data differently.

Lesson 2: Supervised, Unsupervised, and Reinforcement Learning

In this lesson: Supervised Learning & Credit Scoring; Unsupervised Learning & Segmentation;

Key points

- Supervised learning relies on labeled historical data @ "Supervised learning relies on labeled historical data to predict known outcomes"
- Unsupervised learning uncovers hidden client behavioral segments @ "Unsupervised learning analyzes unlabeled data to uncover hidden patterns"
- Reinforcement learning optimizes decisions through dynamic feedback loops @ "Reinforcement learning learns optimal decision-making through dynamic trial, error, and feedback loops"

Examples from the lesson

- I am your instructor, and you are watching course A.I Fundamentals, Module 2 Machine Learning Basics, Lesson 2: Supervised, Unsupervised, and Reinforcement Learning.
- Today, we are taking the next step to compare the three primary engines of artificial intelligence and match each one directly to real financial decisions right here in Trinidad and Tobago.
- The applicant earns TT$ 10,000 in monthly income and carries TT$ 3,000 in existing monthly debt payments.
- The proposed new loan adds TT$ 1,500 in monthly installment payments, bringing the total monthly debt obligations to TT$ 4,500.
- To evaluate risk, the model computes the total debt-to-income ratio by dividing TT$ 4,500 by TT$ 10,000, yielding exactly 45 percent.
- Based on historical repayment data, the supervised model flags any ratio above 40 percent as elevated risk, prompting a closer review before approval.

Lesson 3: When Models Get It Wrong

In this lesson: Overfitting Pitfalls; Training Data Bias; Impact of Bad Inputs; Three Simple Reali

Key points

- AI fails from overfitting, data bias, and bad inputs
- Poor inputs dropped real profit from TT$ 9,000 to TT$ 4,000
- Verify numbers with common sense before trusting AI

Examples from the lesson

- I’m your instructor, and you’re watching "A.I Fundamentals," Module 2: "Machine Learning Basics," Lesson 3: "When Models Get It Wrong."
- To protect your hard-earned money, you need to understand the three main reasons

models fail: overfitting, biased training data, and poor-quality inputs.

- On paper, you enter TT$ 15,000 in expected revenue from sales at the fete.
- You tell the model your raw materials cost TT$ 3,000, booth rental at the venue is TT$ 2,000, and transport via maxi for your team and inventory is TT$ 1,000.
- Based on these poor-quality inputs, the AI adds up your expenses to equal TT$ 6,000.
- It subtracts this TT$ 6,000 total expense from your TT$ 15,000 revenue, predicting a healthy net profit of TT$ 9,000.
- In reality, your actual total expenses come out to TT$ 11,000.
- When you subtract TT$ 11,000 in real expenses from your TT$ 15,000 revenue, your actual net profit drops to just TT$ 4,000.
- Because you trusted bad inputs, you are now short by TT$ 5,000 and struggling to pay that credit union installment.
- Before you ever trust an AI model's output, you must perform three simple reality checks.
- We saw how a small error in input expenses can shrink your predicted profit from TT$ 9,000 down to a real TT$ 4,000, leaving you with a TT$ 5,000 shortfall.

Recap checklist — can you say each one?

Lesson 1: How Machines Learn From Data

Lesson 2: Supervised, Unsupervised, and Reinforcement Learning

☐ Supervised learning relies on labeled historical data @ "Supervised learning relies on labeled historical data to predict known outcomes"

☐ Unsupervised learning uncovers hidden client behavioral segments @ "Unsupervised learning analyzes unlabeled data to uncover hidden patterns"

☐ Reinforcement learning optimizes decisions through dynamic feedback loops @ "Reinforcement learning learns optimal decision-making through dynamic trial, error, and feedback loops"

Lesson 3: When Models Get It Wrong

☐ AI fails from overfitting, data bias, and bad inputs

☐ Poor inputs dropped real profit from TT$ 9,000 to TT$ 4,000

☐ Verify numbers with common sense before trusting AI